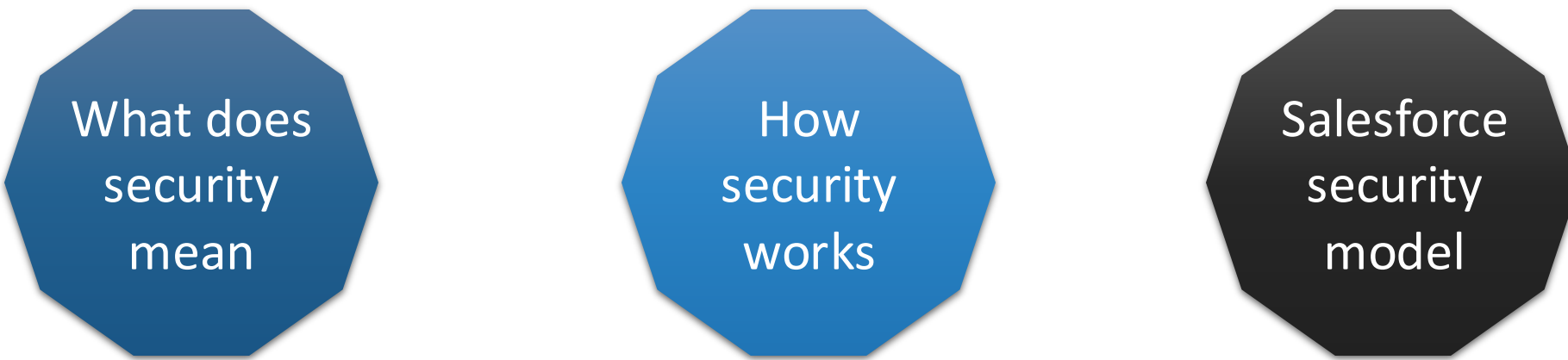


Salesforce Security Model - I



Agenda

- Introduction
- User
- User Authentication
- Auditing
- Profile
- Permission set
- Roles
- Queues



What does
security
mean

How
security
works

Salesforce
security
model

Introduction

- Salesforce provides a comprehensive and flexible data security model to secure data at different levels.
- Salesforce also provides sharing tools to open up and allow secure access to data based on business needs.
- Data is stored in three key constructions
 - Objects
 - Fields
 - Records

Users



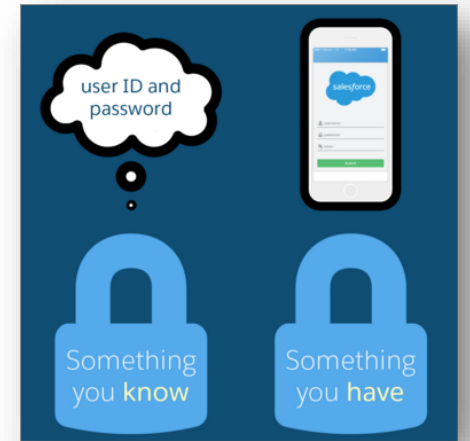
- A user is anyone who logs in to Salesforce.
- Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records.
- Permissions determine a user's ability to:
 - Create / Edit / Delete an object record
 - Empty the organizational Recycle Bin
 - Reset a user's password etc

User Authentication

Authentication means preventing unauthorized access to your organization or its data by making sure each logged in user is who they say are.

- **Multi-Factor Authentication :**

A secure authentication method that requires users to prove their identity by supplying two or more pieces of evidence (or factors) when they log in.



- **Single Sign-On :**

Authentication method that enables users to access multiple applications with one login and one set of credentials.

User Page

User
Omkar Bagwe

User ProfileHelp for this Page

Permission Set Assignments [0] | Permission Set Assignments: Activation Required [0] | Permission Set Group Assignments [0] | Permission Set License Assignments [0] | Personal Groups [0] | Public Group Membership [0] | Queue Membership [3] | Team [1] | Default Account Team [0] | Managers in the Role Hierarchy [0] | OAuth Apps [19] | Third-Party Account Links [0] | Installed Mobile Apps [0] | Authentication Settings for External Systems [0] | Login History [10+] | User Provisioning Accounts [0]

User Detail

Edit | Sharing | Change Password | View Summary

Name	Omkar Bagwe	Role	CEO
Alias	OBagw	User License	Salesforce
Email	omkarbagwe.sfdc@gmail.com [Verified]	Profile	System Administrator
Username	omkar-sfdc-a1@sf.com	Active	☒
Nickname	omkar-sfdc-a1 	Marketing User	☒
Title		Offline User	☒
Company	omkar	Knowledge User	☐
Department		Flow User	☐
Division		Service Cloud User	☒
Address	IN	Site.com Contributor User	☐
Time Zone	(GMT+05:30) India Standard Time (Asia/Kolkata)	Site.com Publisher User	☐
Locale	English (India)	WDC User	☐
Language	English	Mobile Push Registrations	View
Currency	INR - Indian Rupee	Data.com User Type	

Auditing

- Auditing provides information about use of the system
- Salesforce auditing features don't secure the organization by themselves. It should be performed by someone in your organization.
- An auditor can also be referred to as a Salesforce org doctor who conducts thorough inspection and suggests corrective measures to improve the functionality of the platform



Auditing Techniques

- **Record Modification Fields:** Stores the creator and last modifier of a record.
- **Login History:** Lists successful and failed login attempts for 6 months.
- **Field History Tracking:** Tracks changes in selected field values for auditing.
- **Setup Audit Trail:** Logs configuration changes made by administrators.

Profiles

- Profiles define default settings for users. When you create users, you assign a profile to each one.
- Profile is a group/collection of settings and permissions that define what a user can do in Salesforce.
- When you create users, you assign a profile to each one (one user can be assigned with only one profile).

Profile

System Administrator

Users with this profile have the permissions and page layouts listed below. Administrators can change a user's profile by editing that user's personal information.

If your organization uses Record Types, use the Edit links in the Record Type Settings section below to make one or more record types available to users with this profile.

[Login IP Ranges \[0\]](#) | [Enabled Apex Class Access \[98\]](#) | [Enabled Visualforce Page Access \[12\]](#) | [Enabled External Data Source Access \[0\]](#) | [Enabled Named Credential Access \[0\]](#) | [Enabled External Credential Principal Access \[0\]](#) | [Enabled Custom Metadata Type Access \[0\]](#) | [Enabled Custom Setting Definitions Access \[0\]](#) | [Enabled Flow Access \[0\]](#) | [Enabled Service Provider Access \[0\]](#) | [Enabled Custom Permissions \[0\]](#)

Profile Detail [Edit](#) [Clone](#) [View Users](#)

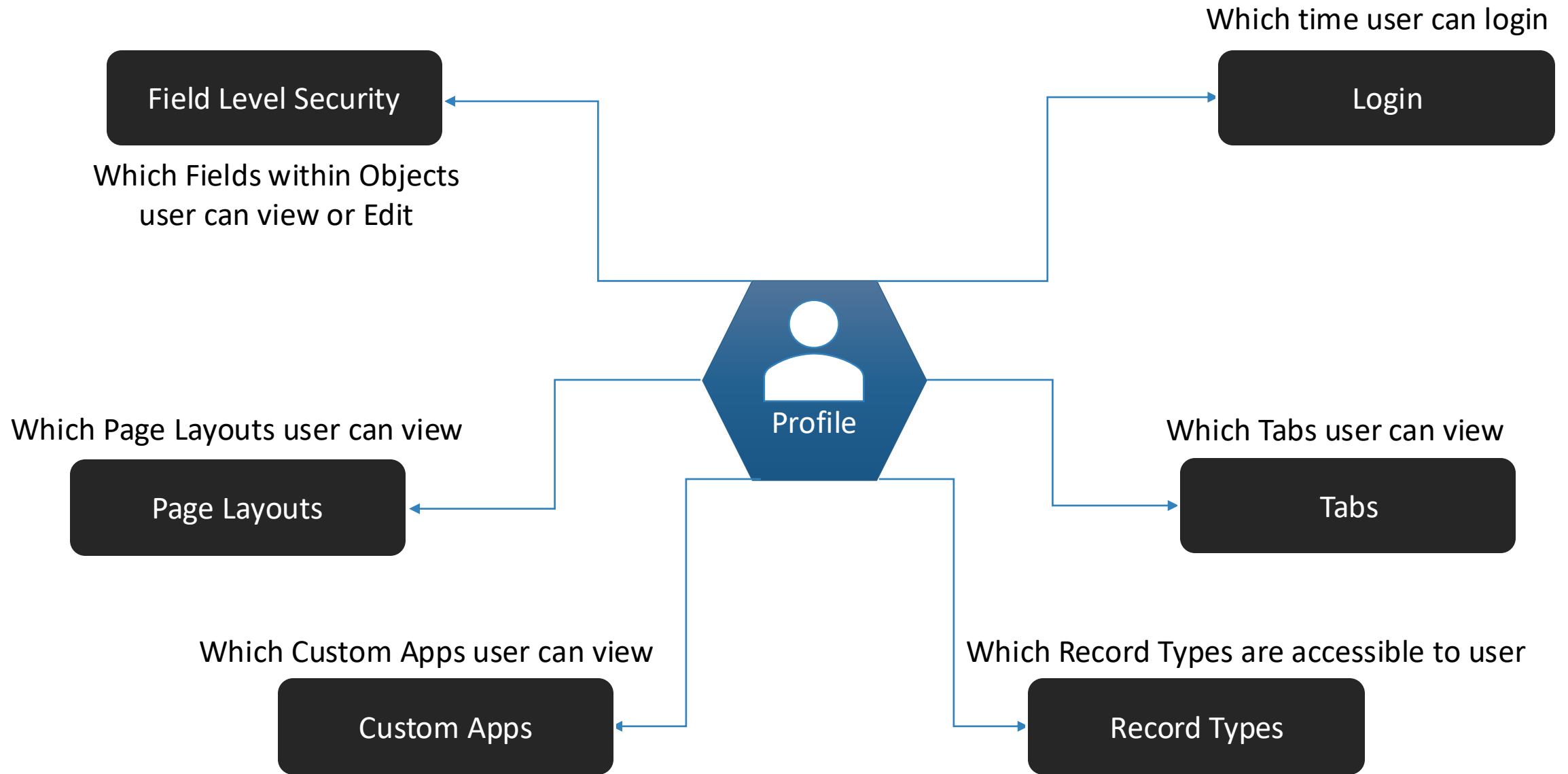
Name	System Administrator		
User License	Salesforce	Custom Profile	☐
Created By	Omkar Bagwe, 09/04/2021, 3:15 pm		Modified By Omkar Bagwe, 24/05/2024, 4:04 pm

Page Layouts

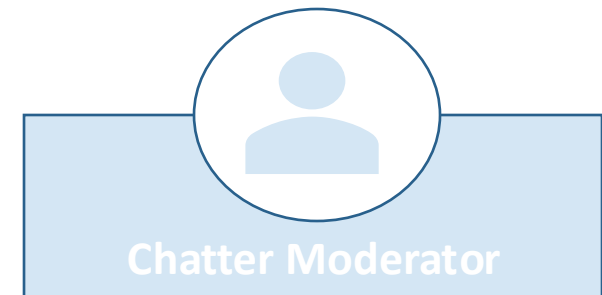
Standard Object Layouts

Global	Global Layout [View Assignment]	Individual	Individual Layout [View Assignment]
--------	--	------------	--

Email Application Not Assigned



Standard Profiles



Permission Set

- Collection of settings and permissions that give users access to various tools and functions.
- Permission sets extend users' functional access without changing their profiles.

Permission Set
Sales Cloud User

Find Settings... | Clone | Manage Assignments | View Summary

Permission Set Overview

Description	Denotes that the user is a Sales Cloud user.	API Name	SalesUserPsi
License	Sales User	Namespace Prefix	force
Session Activation Required	☐	Created By	salesforce.com, inc., 09/04/2021, 3:15 pm
Permission Set Groups Added To	0	Last Modified By	salesforce.com, inc., 11/08/2024, 4:16 pm

Apps

Settings that apply to Salesforce apps, such as Sales, and custom apps built on the Lightning Platform
[Learn More](#)

Object Settings
Permissions to access objects and fields, and settings such as tab availability

App Permissions
Permissions to perform app-specific actions, such as "Manage Call Centers"

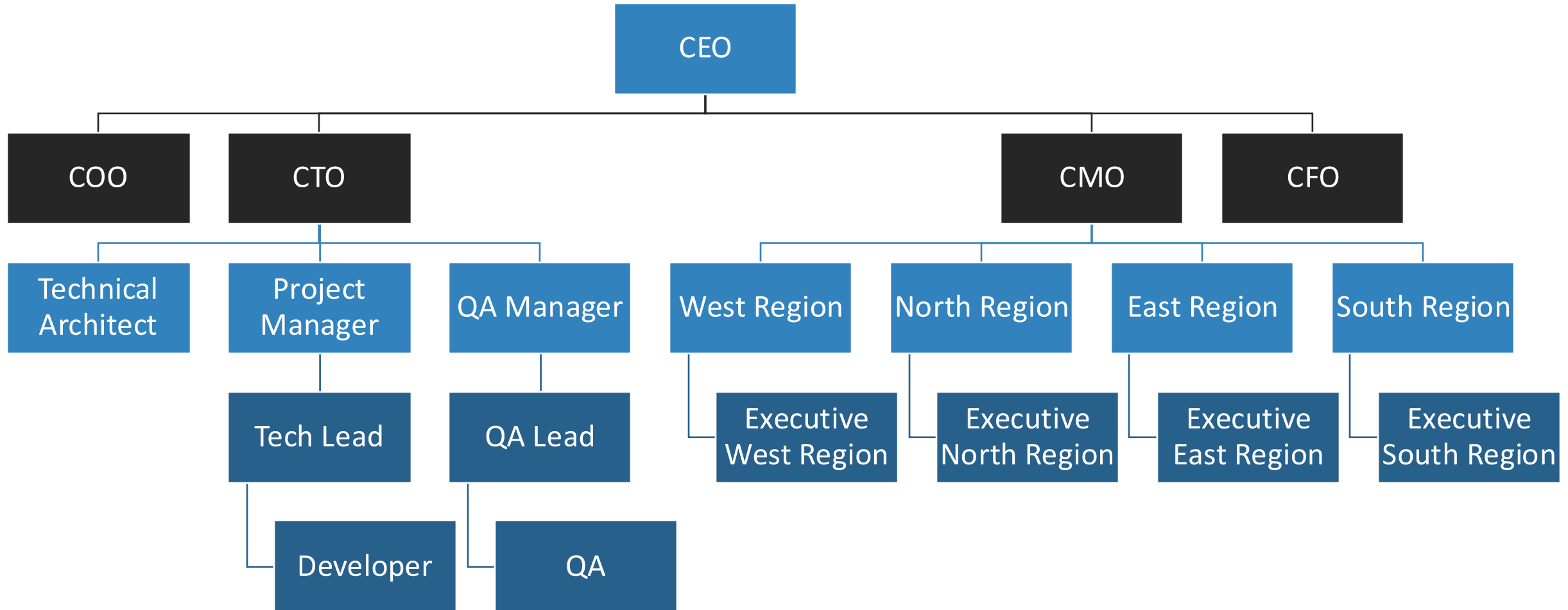
Flow Access
Permissions to execute Flows

External Credential Principal Access
Permissions to authenticate with external credential principal mappings

Custom Metadata Types
Permissions to access custom metadata types

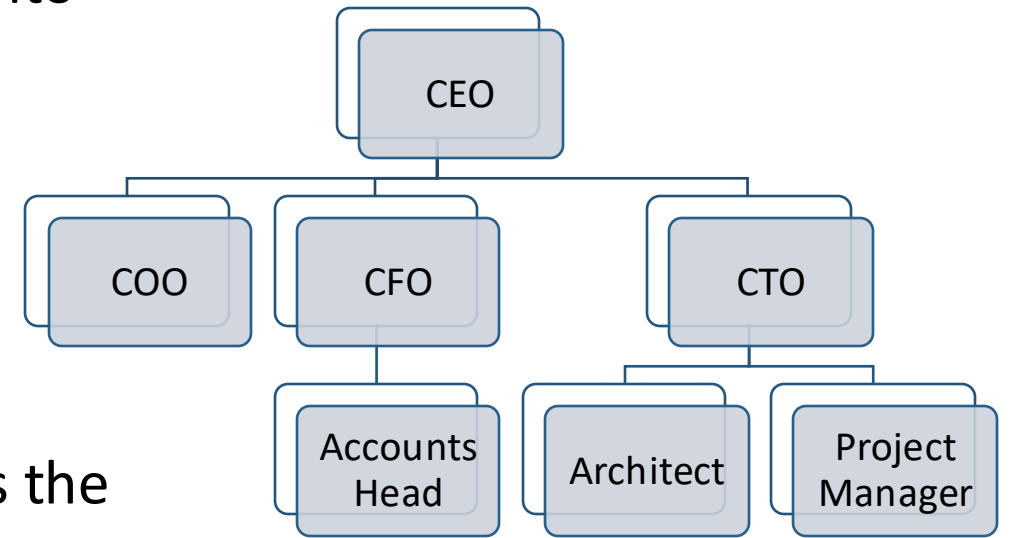
Custom Setting Definitions
Permissions to access custom settings

Organization Chart



Roles

- Role controls the level of visibility that users have into your organization's data.
- Roles control which records are available to Users
- Unlike Profiles
 - No configuration in a Role.
 - Arranged in a hierarchy.
 - Role hierarchy with sharing settings determines the levels of access users have to your Salesforce data.
- Users can access the data of all the users directly below them in the hierarchy.



Roles

Your Organization's Role Hierarchy

S

[Collapse All](#) [Expand All](#)

 **tutorialkart.com**

 [Add Role](#)

 **CEO** [Edit](#) | [Del](#) | [Assign](#)

 [Add Role](#)

 **CFO** [Edit](#) | [Del](#) | [Assign](#)

 [Add Role](#)

 **COO** [Edit](#) | [Del](#) | [Assign](#)

 [Add Role](#)

 **SVP, Customer Service & Support** [Edit](#) | [Del](#) | [Assign](#)

 [Add Role](#)

 **SVP, Human Resources** [Edit](#) | [Del](#) | [Assign](#)

 [Add Role](#)

 **SVP, Sales & Marketing** [Edit](#) | [Del](#) | [Assign](#)

 [Add Role](#)

Queues



- Queues Prioritize, distribute, and assign records to teams who share workloads. Queues help your teams manage leads, cases, service contracts, and custom objects.
- Records once placed in a queue manually or through an automatic case or lead assignment rule, remain there until they're assigned to a user or taken by one of the queue members.
- Any queue member or users above them in the role hierarchy can take ownership of records in a queue.

Architecture of the Salesforce Data and Security Model

- Salesforce stores data in three forms – objects, fields, and records.
- Objects are referred to as tables in databases. Similarly, the fields and records are equivalent to the columns and rows of a table.
- To enforce data layers of security at a personal and organizational level, Salesforce uses three levels of security



Object Level Security

- Defining and enforcing data visibility and data manipulation permissions for Salesforce users.
- It works by controlling access to view, create, edit, or delete records of an object.
- Object-level access can be managed through two configurations - Profiles and Permission Sets.

Object Level Security

Chatter Feeds

Standard Object Permissions

	Basic Access				Data Administration			Basic Access				Data Administration	
	Read	Create	Edit	Delete	View All	Modify All		Read	Create	Edit	Delete	View All	Modify All
Accounts	✓	✓	✓	✓	✓	✓	Goals	✓	✓	✓	✓	✓	✓
Assets	✓	✓	✓	✓	✓	✓	Goal Links	✓	✓	✓	✓	✓	✓
Campaigns	✓	✓	✓	✓	✓	✓	Ideas	✓	✓	✓	✓		
Cases	✓	✓	✓	✓	✓	✓	Leads	✓	✓	✓	✓	✓	✓
Coaching	✓	✓	✓	✓	✓	✓	Macros	✓	✓	✓	✓	✓	✓
Contacts	✓	✓	✓	✓	✓	✓	Metrics	✓	✓	✓	✓	✓	✓
Contracts	✓	✓	✓	✓	✓	✓	Metric Data Links	✓	✓	✓	✓	✓	✓
D&B Companies	✓	✓	✓	✓	✓	✓	Opportunities	✓	✓	✓	✓	✓	✓
Documents	✓	✓	✓	✓	✓	✓	Orders	✓	✓	✓	✓	✓	✓
Duplicate Record Sets	✓	✓	✓	✓	✓	✓	Performance Cycles	✓	✓	✓	✓	✓	✓
Feedback	✓	✓	✓	✓	✓	✓	Price Books	✓	✓	✓	✓		
Feedback Questions	✓	✓	✓	✓	✓	✓	Products	✓	✓	✓	✓		
Feedback Question Sets	✓	✓	✓	✓	✓	✓	Push Topics	✓	✓	✓	✓		
Feedback Requests	✓	✓	✓	✓	✓	✓	Solutions	✓	✓	✓	✓	✓	✓
Feedback Templates	✓	✓	✓	✓	✓	✓	Streaming Channels	✓	✓	✓	✓	✓	✓

Custom Object Permissions

Field Level Security

I want to restrict few sensitive fields from my users. I don't want display these sensitive fields for all users. We can restrict fields by using field level security. We can give read & edit permissions to the field. We can handle these field level permissions in three ways.

1. By using “set field level security” button at field level.
2. By using “view field accessibility” button at field level
3. In profiles/permission sets we can handle field level permissions.

Field Level Security

1. By using set field level security button:

Go to the field you want to enable field level permissions and click on “field level security” button. See the below screen for reference.

Account Field
Annual Revenue
[Back to Account Fields](#)

[Edit](#) [Set Field-Level Security](#) [View Field Accessibility](#)

Field Information

Field Label	Annual Revenue	Field Name	AnnualRevenue
Data Type	Currency(18, 0)		
Help Text			

Field Level Security

2. By using view field accessibility button:

- Go to the field you want to enable field level permissions
- and click on “field accessibility” button.
- then you will navigate to shown screen.

Field accessibility for Field: Annual Revenue

Click on a cell in the table below to change the field's access

	Record Types	
Profiles	Master	Loan Account
ADMIN	Editable	Editable
Contract Manager	Editable	Editable
Cross Org Data Proxy User	Editable	Editable
Custom: Marketing Profile	Editable	Hidden
Custom: Sales Profile	Editable	Editable
Custom: Support Profile	Editable	Editable
Customer Community Login User	Editable	Editable
Customer Community User	Editable	Editable
Customer Portal Manager Custom	Hidden	Hidden
Customer Portal Manager Standard	Hidden	Hidden
Gold Partner User	Editable	Editable
High Volume Customer Portal	Hidden	Hidden
Marketing User	Editable	Editable

Field Level Security

3. Managing Field level security at profile level

Field-Level Security

Standard Field-Level Security

Account	[View]	Opportunity Product	[View]
Asset	[View]	Opportunity Split	[View]
Campaign	[View]	Opportunity Team Member	[View]
Campaign Member	[View]	Price Book	[View]
Case	[View]	Price Book Entry	[View]
Contact	[View]	Product	[View]
Contract	[View]	Social Persona	[View]
Event	[View]	Solution	[View]
Idea	[View]	Task	[View]
Lead	[View]	User	[View]
Opportunity	[View]		

Custom Field-Level Security

- To handle Field permissions, go to profile you want, choose and go to Field level security
- section and search for object you want enable field permissions, click on view and
- edit and provide permissions for required fields and save.

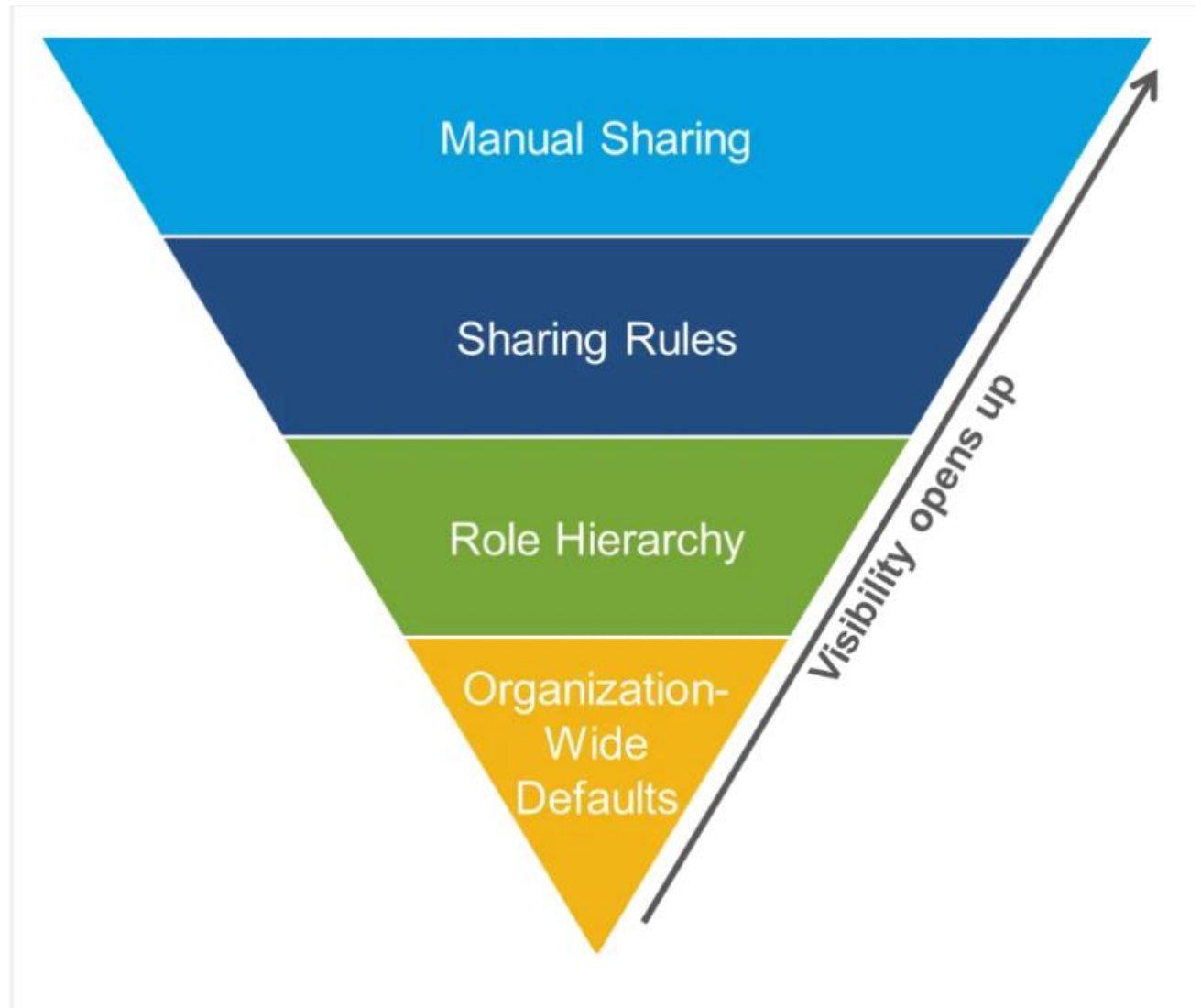
Record Level Security

- Feature that ensures the right data is accessible to the right people
- This plays important role in preserving data confidentiality and maintaining its authenticity.



Ways to share records...

- Salesforce provides five ways to share records with others and access others' records.
- Start by configuring org-wide defaults, to lock down your data to the most restrictive level. Then you use the other record-level security tools to grant additional access to selected users when needed.



Any Questions ?



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B
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Thank You



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